Worksheet W7 · W-Seminar "Time"

Refine and evaluate

Test your instrument in a pilot, read the signals – and say what your data can and cannot show.

1 Read the pilot: a time-pressure scale

24 students answered six statements from 1 (strongly disagree) to 5 (strongly agree); tp3 is already reversed. **α with all six items = 0.71.**

	statement	mean	SD	item–rest r	α without it
tp1	I often feel I don't have enough time for my schoolwork.	2.96	1.04	0.58	0.63
tp2	My days feel rushed.	3.08	1.02	0.70	0.59
tp3	I usually have time to relax in the evening. (reversed)	3.08	1.21	0.49	0.67
tp4	I feel stressed and tired in the morning.	3.38	0.71	−0.18	0.80
tp5	Deadlines often catch me by surprise.	3.33	1.27	0.72	0.57
tp6	I sometimes have more to do than I can manage.	4.58	0.50	0.44	0.70

tp6: 10 students ticked 4, 14 ticked 5, nobody ticked 1, 2 or 3.

a) I would drop or split item *tp4* because

it asks two things at once – its item–rest r is even negative

b) Without it, α rises to *0.80*. c) tp6 shows a *ceiling effect*.

Better wording for tp6: *Most days I have more to do than I can manage.*

2 More signals one per data type

Coding: two coders agree on 77 % of the lines, by chance 20 %. κ = (0.77 − 0.20) / (1 − 0.20) = *0.71*

Good enough to code the whole corpus? *no – aim for 0.80: add a decision rule to the codebook*

Test: item q4 is solved by 97 % of the students: *too easy – it separates nobody*

Item q7 has a *negative* discrimination. First check: *the key (a second correct answer?)*

Experiment: two estimates of a 3-minute wait are 210 and 165. *seconds – rewrite the instruction*

The fast music was rated 3.6, the slow music 3.3 (1–5): *manipulation too weak – stronger contrast*

3 My pilot plan it, run it, log every change

Who, how many, when: *5 students from Year 9, next Tuesday*

I watch ☒ time needed ☒ questions they ask ☐ think-aloud ☒ impossible values
☐ α / κ / item analysis ☒ manipulation check ☐ other: _____

Pair 3 min Tell your partner what is most likely to go wrong in your pilot. Note their idea: *people know the song – ask afterwards and note it*

what I saw	what I changed	why
2 of 5 asked "minutes or seconds?"	*instruction with example "2 min 40 s"*	*no more impossible values*

4 Which analysis answers the question?

Choose: **compare** groups · **relate** two variables · track a **change** · **count** categories.

research question	analysis	chart	key number
Do evening types sleep less on school nights?	*relate*	*scatter plot*	*r (or ρ)*
Does fast music make a wait feel shorter than slow music?	*compare*	*box plot*	*d and p*
Has the share of "gonna" in song lyrics grown since the 1960s?	*count*	*line chart (shares)*	*% per decade*
Did a grammar lesson improve test scores more than in a control class?	*change*	*box plot of gains*	*mean gain, d*
My question (version 4)**:** *music tempo and a 3-min wait*	*compare*	*box plot*	*d and p*

Share 1 min each Explain your analysis plan to your partner: which result would surprise you – and what would you do then?

5 Read the output and write it up

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fast:  M = 2.60 min (SD 0.61), n = 28   slow:  M = 3.04 min (SD 0.62), n = 28
difference 0.45 min, d = 0.73   permutation test:  p = 0.009
```

a) The effect is *medium to large* (d ≈ 0.2 small · 0.5 medium · 0.8 large).

b) p = 0.009 means, in your own words:

If the music made no difference, a gap this large would appear in fewer than 1 in 100 shuffles.

c) Your results sentence (numbers, no interpretation yet):

Students with fast music estimated the wait as shorter (M = 2.60, SD = 0.61) than students with slow music (M = 3.04, SD = 0.62); d = 0.73, p = .009.

d) One limitation:

One school, volunteers only – and we measured estimated, not felt, time.

6 Found nothing? Diagnose before you conclude

A pilot with 12 students: screen time before bed and time pressure, **r = 0.04**. Think 2 min Pair 3 min Find three reasons why r could be close to 0 although there is a link – and decide what to do.

1 *too few people (n = 12) – r jumps around*

2 *unreliable measure, e.g. one vague item*

3 *restricted range / U-shape / subgroups cancel out – plot it*

In the pilot I would: *fix the instrument (more items, wider sample), pilot again*

In the main study I would: *report r, n and effect honestly; discuss the causes*

7 My question – version 4 question in box 4

spark → question → working thesis → **testable claim** → thesis

Testable claim after the pilot – what changed since W6, and why? *clearer tempos, 64 per group*

Draft thesis statement, if the results go as expected: *Fast music makes a 3-minute wait feel shorter.*